UNIT 3

Exercise 7: Domain-specific chatbot using LangChain and a vector database

Aim

To generate numerical embeddings for text and identify semantically similar sentences using embedding similarity.

Algorithm

- Start the program.
- Load a pretrained sentence-embedding model.
- Define a collection of text documents.
- Convert the documents into numerical vectors.
- Convert the user's query into an embedding.
- Calculate similarity scores between the query and documents.
- Display the most similar document.
- Stop.

Python Code and Output

```
from langchain_core.documents import Document
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_community.vectorstores import Chroma

# Domain: Artificial Intelligence
knowledge = [
    "Artificial intelligence creates systems that perform intelligent tasks.",
    "Machine learning trains models using data.",
    "Deep learning uses multilayer neural networks.",
    "NLP enables computers to process human language.",
    "RAG combines document retrieval with answer generation."
]

lc_documents = [Document(page_content=t) for t in knowledge]

lc_embeddings = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)

db = Chroma.from_documents(
    documents=lc_documents,
    embedding=lc_embeddings,
    collection_name="ai_lab_knowledge"
)

question = "What is machine learning?"

results = db.similarity_search(question, k=3)
context = "\n".join(d.page_content for d in results)
```

```
Question: What is machine learning?
Retrieved documents:
- Artificial intelligence creates systems that perform intelligent tasks.
- Deep learning uses multilayer neural networks.
- NLP enables computers to process human language.
Chatbot answer: Deep learning
```

Result

Text embeddings were generated successfully, and semantic similarity search identified the most relevant document.